This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

# Robust Spiking Neural Networks by Temporal Mutual Information

Mengting Xu[1,2], Shi Gu[1,2], Peng Lin[1,2], De Ma[1,2], Huajin Tang[1,2], Qian Zheng[1,2]*, Gang Pan[1,2]
[1]The State Key Lab of Brain-Machine Intelligence, Zhejiang University
[2]College of Computer Science and Technology, Zhejiang University
{xumengting, gus, penglin, made, htang, qianzheng, gpan}@zju.edu.cn

## Abstract

*Spiking Neural Networks (SNNs) have attracted increasing attention for their biologically inspired temporal dynamics. As their applications expand, understanding their robustness has become an important research focus. However, little is known about how the intrinsic temporal characteristics of SNNs affect robustness. In this work, we revisit SNN robustness from an information-theoretic perspective and reveal the pivotal role of temporal characteristics. We establish a theoretical link between robustness error and the mutual information (MI) between inputs and latent representations along the temporal dimension, grounded in the information bottleneck principle. Through an analysis of spike-based information transmission, we show that temporal characteristics inherently compress MI, thereby tightening the robustness error bound. Building on this insight, we propose a Temporal Mutual Information (TMI) regularizer that explicitly exploits temporal characteristics to enhance robustness. Extensive experiments on CIFAR-10, CIFAR-100, DVS-CIFAR10, Tiny-ImageNet, and ImageNet demonstrate that our method consistently improves SNN robustness across various architectures and attack settings. Code is available at* https://github.com/zju-bmi-lab/SNN_TMI_code.

## 1. Introduction

Spiking neural networks (SNNs) have garnered increasing attention in recent years due to their unique temporal dynamics and energy-efficient computing mechanisms [5, 18–20, 28, 45]. By mimicking the sparse and event-driven nature of biological neurons, SNNs have shown promising performance in a wide range of tasks, particularly in the neuromorphic computing [36], low-power edge applications [7, 46], and real-time sensory processing [29, 47]. These distinctive properties allow SNNs to outperform the traditional artificial neural networks (ANNs) in scenarios requiring temporal information integration and low-latency responses, making them a compelling alternative for next-generation intelligent systems.

With the expanding adoption of SNNs, concerns about their robustness, particularly in adversarial perturbations [30], have drawn significant attention [34, 35]. Existing efforts to enhance SNN robustness can be broadly categorized into two directions. The first line of work focuses on the spatial features of SNNs, drawing heavily from established ANN techniques such as adversarial training [23], input perturbation [14], and weight regularization [12]. However, these methods often overlook the distinctive temporal characteristics of SNNs, treating them as static and thereby limiting their effectiveness. The second direction leverages the temporal dynamics inherent in SNNs, exploring techniques like temporal frequency-based encoding [44], and temporal-based gradient computation [27] to better exploit temporal information. While these approaches improve temporal information modeling, they rarely examine how or why temporal characteristics influence SNN robustness.

In this work, we demonstrate that incorporating temporal characteristics into the training of spiking neural networks (SNNs) significantly improves robustness compared to methods that rely solely on spatial features. We first establish a theoretical connection between model robustness error and the mutual information between inputs and internal representations along the temporal dimension, grounded in the information bottleneck principle [39]. We then show that temporal characteristics inherently reduce mutual information by analyzing spike-based information transmission mechanisms unique to SNNs [43], i.e., the cumulative firing mechanism and spike temporal dependency, thereby yielding tighter bounds on robustness error. Building on this insight, we propose a temporal mutual information (TMI) regularization that complements conventional spatial loss functions by explicitly minimizing mutual information in the temporal domain. This joint regularization improves the robustness of SNNs against adversarial perturbations. Our main contributions are:

- We theoretically show that the upper bound of the robust-

*Corresponding author.

ness error in deep neural networks is governed by mutual information between the input and the latent representation, following the information bottleneck principle.
- By analyzing the spike-based information transmission mechanisms unique to SNNs, we show that temporal characteristics inherently induce lower mutual information compared to spatial features alone, thereby enabling a more compact upper bound on model robustness error and enhancing overall robustness.
- We introduce a mutual information estimation method on temporal characteristics and a temporal mutual information (TMI) regularizer tailored for SNNs. Extensive experiments on CIFAR-10, CIFAR-100, DVS-CIFAR10, Tiny-ImageNet and ImageNet demonstrate that temporal characteristics are critical to improving SNN robustness, and TMI regularizer achieves state-of-the-art performance under various noisy conditions.

## 2. Related Work

Existing efforts to enhance the robustness of spiking neural networks (SNNs) can be broadly categorized into two directions. The first focuses on the spatial features of SNNs, often adapting techniques from artificial neural networks (ANNs). One of the most widely used empirical defenses is adversarial training (AT) [38], which demonstrates that incorporating adversarial examples into the training set can improve model robustness against attacks. HIRE-SNN improves the efficiency of adversarial training in SNNs by partitioning the total training time steps $T$ into $N$ equal-length intervals [23]. SNN-RAT introduces a regularized adversarial training scheme with low computational overhead [12], while MPPD integrates membrane potential perturbation dynamics into training [14]. However, the use of surrogate gradients in SNNs reduces the effectiveness of adversarial training compared to its application in ANNs [23]. More critically, adversarial training does not provide intrinsic robustness, as it depends on external adversarial examples and can increase the model's vulnerability to other types of attacks [37]. In addition, these methods largely neglect the temporal characteristics of SNNs, limiting their effectiveness in enhancing robustness. The second direction leverages the temporal dynamics inherent to SNNs. FEEL-SNN proposes a frequency-based encoding scheme in the temporal domain to better exploit temporal information [44]. Gradient sparsity regularization addresses optimization along the temporal dimension [14], and [13] shows that spike timing coding can enhance robustness against adversarial attacks. Although these methods improve temporal information modeling, they seldom investigate how or why temporal properties influence SNN robustness.

In contrast to prior work, this paper shows that temporal characteristics inherently induce lower mutual information compared to spatial features alone, thereby enabling a more compact upper bound on model robustness error and enhancing overall robustness. Additional related work on the information bottleneck principle and mutual information in the context of adversarial attacks is provided in the Appendix.

## 3. Preliminary

### 3.1. Spike Neuron

The membrane potential $\boldsymbol{U}$ of the Leakly-and-Fire (LIF) neuron in SNN updated as [43]:

$$\boldsymbol{U}(t+1) = \tau \boldsymbol{U}(t) + \boldsymbol{I}(t), \tag{1}$$

where $\tau$ is the constant leaky factor, $\boldsymbol{U}(t)$ is the membrane potential at time $t$, and $\boldsymbol{I}(t)$ denotes the pre-synaptic current input, which is the product of synaptic weight $\boldsymbol{W}$ and spiking input $\boldsymbol{S}(t)$. The firing function and hard reset mechanism of LIF are described as follows

$$\boldsymbol{S}(t+1) = \mathcal{H}(\boldsymbol{U}(t+1) - V_{th}), \tag{2}$$

$$\boldsymbol{U}(t+1) = \boldsymbol{U}(t+1) \cdot (1 - \boldsymbol{S}(t+1)), \tag{3}$$

where $\mathcal{H}(\cdot)$ represents the Heaviside step function. The neuron fires a spike $S(t)$ at time $t$ if and only if $\boldsymbol{U}(t)$ exceeds the threshold $V_{th}$, and then the membrane potential $\boldsymbol{U}(t)$ reset to 0. The output spike $\boldsymbol{S}(t+1)$ will become the post synaptic spike and propagate to the next layer.

### 3.2. Information Bottleneck Principle

The Information Bottleneck (IB) [39, 40], presents a systematic approach for discovering a latent representation $Z$ that encapsulates the information inherent in the input variable $X$ to the output $Y$. The mutual information of $X$ and $Z$ can be written as $I(X;Z)$. $I(X;Z) = D_{KL}(\mathbb{P}_{XZ} || \mathbb{P}_X \otimes \mathbb{P}_Z)$, where $D_{KL}$ is the Kullback-Leibler divergence [41], $\mathbb{P}_{XZ}$ is the joint distribution and $\mathbb{P}_X \otimes \mathbb{P}_Z$ is the product of the marginal distributions of input variables $X, Z$, respectively. In essence, the Information Bottleneck principle not only quantifies the extent to which $Z$ compresses information from $X$ through $I(X;Z)$ but also gauges how effectively $Z$ predicts $Y$ through $I(Z;Y)$. In practice, the IB principle is realized by minimizing the following IB Lagrangian:

$$\min\{I(X;Z) - \beta \cdot I(Z;Y)\}, \tag{4}$$

where $\beta$ represents a positive parameter, used to control the trade-off between compression and prediction.

## 4. Method

### 4.1. Upper Bound of Robustness Error

The neural networks can be considered as a Markov chain of successive representations of the input where information flows obeying the structure [2]: $X \to Z \to Y$, where

$X$ is the input data, $Z$ is the latent representation, and $Y$ is the output. And by the data processing inequality [8], we have $I(X;Z) \geq I(X;Y)$, where $I(X;Z)$ represents the mutual information between $X$ and $Z$. For a trained model, the $I(Z;Y)$ term can be used to measure the performance of the model in a supervised learning setting. Then the generalization error can be presented as, following the bound derived in Theorem 4 of Shamir et al. [33] :

$$\begin{aligned} |I(Z;Y) - \hat{I}(Z;Y)| \leq C(c_1 \log(m)\sqrt{|\mathcal{Z}|I(X;Z)} \\ + c_2|\mathcal{Z}|^{\frac{3}{4}} I(X;Z)^{\frac{1}{4}} + c_3 \hat{I}(X;Z)) \end{aligned}, \quad (5)$$

where $C = \sqrt{\frac{c \log(|\mathcal{Y}|/\delta)}{m}}$, $\delta$ is a confidence parameter with $\delta \in (0,1)$, $c$ is a small constant, and $c_1, c_2, c_3$ depend only on marginal probability distributions $p(x)$ and $p(y)$. $m$ is the sample size drawn from $p(x,y)$. And $\mathcal{Z}$ is the space of $Z$. $\hat{I}(Z;Y)$ is the empirical estimates of $I(Z;Y)$, $\hat{I}(X;Z)$ is the empirical estimates of $I(X;Z)$.

**Derive the mutual information as upper bound of the robustness error.** To establish the relationship between model robustness and mutual information, in this part, we analyze the impact of adversarial examples. We denote $X+\epsilon$ as the adversarial example with perturbation $\epsilon$, and $Z+\epsilon'$ as the corresponding latent representation. When in the ideal state of the robust model, based on the information flow, compared with $Z$, $Z+\epsilon'$ can be regarded as a lower-dimensional representation of the network for it contains more redundant noise [48]. Then the information flow can be written as $X+\epsilon \to Z+\epsilon' \to Z \to Y$. According to [8, 48], $H(Y|Z)$ represents the uncertainty of random variable $Y$ when given $Z$, where $H(\cdot)$ is the Shannon entropy [8, 26]. In the inference phase, the model parameters $\theta$ are fixed, and there is no uncertainty of $Y$ when we know $Z$ or $Z+\epsilon'$ in the inference phase. Therefore, $\hat{H}(Y|Z) = 0$ and $\hat{H}(Y|Z+\epsilon') = 0$. After combining all of these together, we have

$$\begin{aligned} \hat{I}(Z+\epsilon';Y) &= \hat{H}(Y) - \hat{H}(Y|Z+\epsilon') = \hat{H}(Y) \\ &= \hat{H}(Y) - \hat{H}(Y|Z) \\ &= \hat{I}(Z;Y), \end{aligned} \quad (6)$$

where $\hat{I}(Z+\epsilon';Y)$ is the empirical estimates of $I(Z+\epsilon';Y)$, $\hat{H}(Y|Z)$ is the empirical estimates of $H(Y|Z)$. Therefore, based on Eq. (5), the robustness error of the model can be formulated as

$$\begin{aligned} |I(Z;Y) - \hat{I}(Z+\epsilon';Y)| &= |I(Z;Y) - \hat{I}(Z;Y)| \\ &\leq C(c_1 \log(m)\sqrt{|\mathcal{Z}|I(X;Z)} \\ &+ c_2|\mathcal{Z}|^{\frac{3}{4}} I(X;Z)^{\frac{1}{4}} + c_3 \hat{I}(X;Z)). \end{aligned} \quad (7)$$

The Eq. (7) illustrates the relationship between the robustness error (as embodied in the difference between $\hat{I}(Z+\epsilon';Y)$ and $I(Z;Y)$), and the mutual information $I(X;Z)$, where we can obtain that the upper bound of the robustness error for a deep neural network is determined by mutual information between the input feature $X$ and its latent representation $Z$.

### 4.2. Temporal Characteristics in SNNs Facilitate Obtaining a Compact Upper Bound

According to Eq. (7), to better constrain the robust error of the model, we need to calculate more compact mutual information between the input $X$ and the latent representation $Z$ as much as possible. And the mutual information $I(X;Z)$ can be formulated as [2, 8]:

$$I(X;Z) = H(X) - H(X|Z), \quad (8)$$

where $H(\cdot)$ is the Shannon entropy, $H(X|Z)$ is the conditional entropy, which represents the uncertainty of $X$ when given $Z$ [48]. Therefore, given $X$, *i.e.*, the value of $H(X)$ is fixed, we can measure the compact of mutual information by the uncertainty $H(X|Z)$. Based on this analysis, in the next, we show that mutual information based on temporal characteristics increases the uncertainty of $H(X|Z)$ compared to spatial features alone, thus facilitating obtaining a compact mutual information upper bound.

**Mutual information calculation only based on spatial feature.** On the one hand, since mutual information measures the correlation between two distributions, calculating mutual information in high dimensions on the spatial feature of a single image in ANN is challenging. Therefore, many researches focus on estimating the mutual information by converting a single image into a distribution by sampling from batches or adding random noises [4, 32]. However, these methods destroy the specificity of the image since the labels of the samples in the batch are different and will lead to a decrease in the accuracy of downstream tasks. On the other hand, when considering the mutual information calculation only based on spatial features, we assume that the input spatial feature $X^S$ and the latent spatial feature $Z^S$ take values in the distributions $\mathcal{X}^S$ and $\mathcal{Y}^S$ respectively, then the mutual information $I(X^S;Z^S)$ based on spatial feature can be represented as:

$$I(X^S;Z^S) = H(X^S) - H(X^S|Z^S), \quad (9)$$

where $H(X^S|Z^S)$ is the conditional entropy, which represents the uncertainty of $X^S$ when given $Z^S$. In the inference phase of ANNs, according to [2], the one-to-one mapping of the former feature to the latter feature (as shown in Fig. 1(a)) can be described as:

$$Z^S = g(X^S) = \sigma(\boldsymbol{W}X^S + b), \quad (10)$$

where $g(\cdot)$ is the mapping function, $\boldsymbol{W}$ is the weight of the model, $b$ is the bias, and $\sigma(\cdot)$ is the activation function.

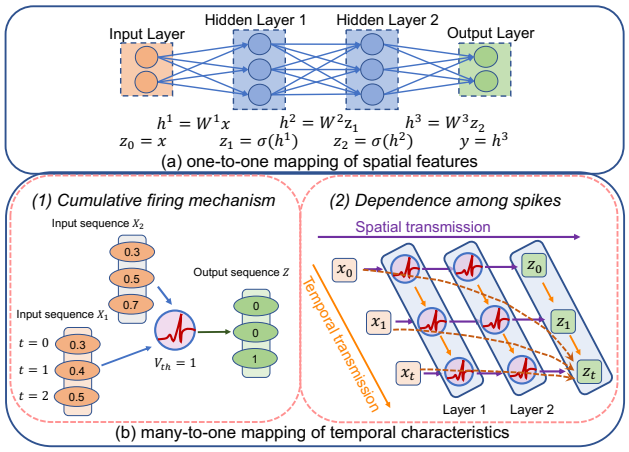



Figure 1. Different mapping between spatial features and temporal characteristics. (a). one-to-one of spatial features on ANNs. (b). many-to-one of temporal characteristics on SNNs.

And since the model is deterministic with the fixed parameters in the inference phase, we can obtain the $X^S$ according to the Eq. (10) when we know $Z^S$, *i.e.*, there is no uncertainty of $X^S$ when we know $Z^S$ [2, 48]. Therefore, we have $H(X^S|Z^S) = 0$ when calculating mutual information based on spatial features alone. However, this results in relatively loose mutual information. According to Eq. (7), the more compact the mutual information, the better its constraint on model inherent robustness. Therefore, calculating more compact mutual information based on appropriate feature encoding is crucial.

**Mutual information calculation based on temporal characteristics incorporation.** In contrast, SNNs offer a unique opportunity to calculate mutual information based on temporal characteristics [17], where a spike train can be regarded as $T$ data sampled from the distribution, $T$ is the time step, thus alleviating the above limitations. We assume that the input temporal characteristics $X$ and the latent temporal characteristics $Z$ take values in the distributions $\mathcal{X}$ and $\mathcal{Z}$ respectively. Different from mutual information calculation based on spatial features as analyzed above, in SNNs, we can regard $X \rightarrow Z$ as a many-to-one mapping (as shown in Fig. 1(b)). We can get this point from two aspects. First and foremost, temporal characteristics emerge through the **cumulative firing mechanism** inherent in spiking neurons. This signifies that the accumulated membrane potential of the presynaptic input $X$ triggers a spike $Z$ as long as it crosses the threshold. Consequently, during the backpropagation process, even if the model parameters are known, we cannot recover the specific membrane potential value from the spike (as in the case of "$X_1$" and "$X_2$" shown at the left of Fig. 1(b)). Therefore, when calculating mutual information based on temporal characteristics, even if $Z$ is known, the uncertainty of $X$ persists, *i.e.*, $H(X|Z) > 0$. Secondly, there is a **dependence among spikes in the spike train**. The characteristic of the current time step in a spike train is influenced by the residual membrane potential from the preceding time step. That is to say, even if we know a specific spike $z_t$ in $Z$, we cannot get the specific $x_t$ in $X$, because $z_t$ is not only affected by $x_t$, but also affected by $x_i$, $i \in \{1, ..., t-1\}$ (as shown at the right of Fig. 1(b)). Therefore, the uncertainty of $X$ persists, and $H(X|Z) > 0$. Then, if we assume the input $X^S = X$, acording to Eq.(8), we can obtain $I(X;Z) < I(X^S;Z^S)$, since $H(X|Z) > H(X^S|Z^S) = 0$.

Even though as pointed out in [2] we can add absolutely continuous noise to the internal spatial features to obtain the stochastic network. However, this requires extremely high requirements for adding noise, for it may cause a decrease in $I(Z|Y)$ in IB principle, thus affecting the discriminative ability of the network. Through the above analysis, MI based on temporal characteristics will be more compact than that based on spatial features, so it can better constrain the inherent robustness of the model according to Eq. (7). It also reflects the necessity of the temporal characteristics of SNN to inherent robustness.

### 4.3. Robust SNNs with the TMI Regularizer

**Mutual information calculation in SNNs.** Gribkova *et al*. use the histogram partition method to calculate mutual information from spike trains in a thalamocortical network [17], where they bundle spikes together in bins to calculate the statistics. However, the histogram partition method ignores the continuous information representation contained between spikes. Besides, SNNs pursue fewer time steps to improve efficiency, but when time step is small, the distribution represented by a few histogram bins can hardly provide discriminative information. To solve these problems, in this study, we calculate the average pixel value at each time step in temporal characteristics (Fig. 2 (b)), and use kernel density estimator (KDE) to present a continuous information representation for spike trains [31] which is desirable to estimate a continuous distribution to estimate statistics of spike firing rate (Fig. 2 (c)). It is much smoother than histogram and provides faster convergence toward the true distribution [31]. Specifically, as shown in Fig. 2 (b), to eliminate the influence of possible noise in the spatial features, as well to obtain more comprehensive spatial information in spike trains, we average the channel (Eq. (12)), height, and width dimensions (Eq. (13)), then we get the 1-dimension spike firing rate trains $X$ and $Z$. The detailed formulas are as follows:

$$X_{input} \in \mathcal{R}^{T \times C_i \times H_i \times W_i}, \tag{11}$$

$$\hat{X}_{input} \in \mathcal{R}^{T \times H_i \times W_i} = \frac{1}{C_i} \sum_{j=C_1}^{C_i} x_i[:, j, :, :], \tag{12}$$

$$X \in \mathcal{R}^{T} = \frac{1}{H_i \times W_i} \sum_{j=H_1}^{H_i} \sum_{q=W_1}^{W_i} X_i[:, j, q], \tag{13}$$

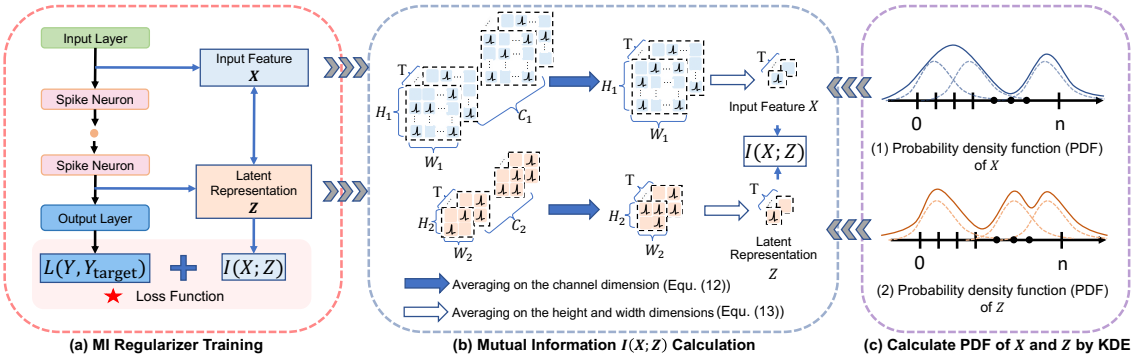



Figure 2. TMI regularizer based on SNN temporal characteristics. (a) TMI regularizer $I(X;Z)$ is calculated between input feature $X$ of the first layer and the related latent representation $Z$ of the last spike neuron. (b) Calculate the average spike firing at each time step. (c) Calculate the probability density function of the temporal characteristics by KDE [31].

where $X_{input}$ is the temporal characteristic of the input layer, $T \times C_i \times H_i \times W_i$ is the shape of $X_{input}$, $T$ is the time step, $\hat{X}_{input}$ is the temporal characteristic after channel averaging, and $X$ is the final 1-dimension spike firing rate train. The latent firing rate train $Z$ is obtained as the same way by replacing $T \times C_i \times H_i \times W_i$ with its shape $T \times C_l \times H_l \times W_l$.

Let $\{x_1, ..., x_T\} \in [0,1]$, $\{z_1, ..., z_T\} \in [0,1]$ represent the final 1-dimension spike firing rate train of $X$ and $Z$, respectively. Then we utilize KDE to calculate the probability density function (PDF) of variables $X$ and $Z$ as follows:

$$PDF_X = \frac{1}{T}\sum_{i=1}^{T}\exp(-\frac{(x_i - bx_j)^2}{2\sigma^2}), bx_j \in bx_1, ...bx_n, \tag{14}$$

$$PDF_Z = \frac{1}{T}\sum_{i=1}^{T}\exp(-\frac{(z_i - bz_j)^2}{2\sigma^2}), bz_j \in bz_1, ...bz_n, \tag{15}$$

where $T$ is the time step, $n$ represents the number of state bins of variable $X$ and $Z$. $bx_j$ and $bz_j, j \in 1, ..n$ are the center value of $n$ bins of variable $X$ and $Z$, $\sigma$ is the parameter of the gaussian kernel function [31]. Then the marginal probability distribution $(p(bx), p(bz))$ and the joint probability distribution $p(bx, bz)$ of $X$ and $Z$ can be obtained by Eq. (14) and Eq. (15). And the mutual information $I(X;Z)$ is formally defined as

$$I(X;Z) = \sum_{i=1}^{n}\sum_{j=1}^{n} p(bx_i, bz_j) log(\frac{p(bx_i, bz_j)}{p(bx_i)p(bz_j)}). \tag{16}$$

**Temporal mutual information (TMI) regularizer construction.** The theoretical analysis above provide us the motivation for exploiting $I(X;Z)$ as a regularization term for model robustness enhancement. Therefore, we embed mutual information into the training process of the model combined with the conventional spatial loss, by reducing the mutual information between the input and its latent temporal characteristics, thus improving the SNN inherent robustness to any attack type. The overall objective function of the TMI regularizer is described as:

$$\mathcal{L} = \mathcal{L}_{base}(Y, Y_{target}) + \lambda I(X;Z), \tag{17}$$

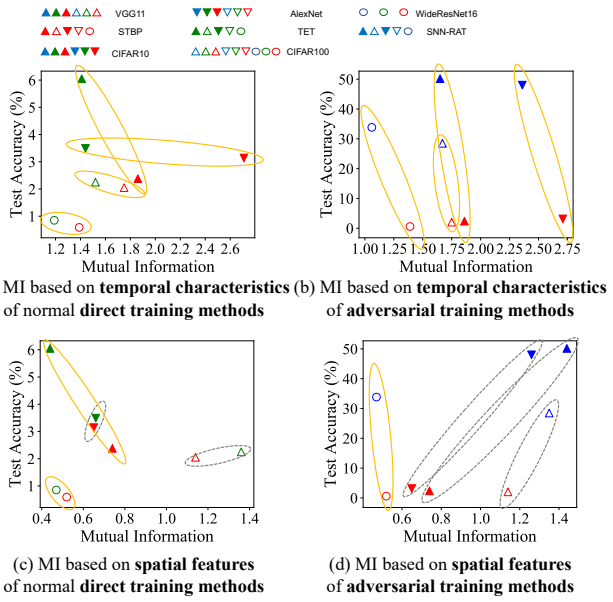
where $\mathcal{L}_{base}(\cdot)$ represents any loss function, such as the commonly used cross-entropy loss in spatial features of SNNs. $\lambda$ is a tunable scaling parameter which is fixed.

**MI regularizer implementation.** It is important to select a reasonable and representative latent representation $Z$ from different layers. Let $Z$ represent the feature of the deep spike neuron layer except for the classification layer, and denote $Z'$ the feature of the shallow spike neuron layer, we have $I(X;Z) \leq I(X;Z')$. If we choose the latent representation from shallow layer, the higher $I(X;Z')$ indicates that extraction ability of this layer is poor. This will affect the final feature extraction ability of the model. Therefore, in TMI regularizer, as shown in Fig. 2 (a), we calculate MI between the input feature $X$ of the first layer and the related latent representation $Z$ of the last spike neuron layer.

## 5. Experiments

### 5.1. Experimental Setup

We use CIFAR-10, CIFAR-100 [21], DVS-CIFAR10 [25], Tiny-ImageNet, and ImageNet [10] for evaluation. The networks are VGG11, AlexNet [22], and VGGSNN [11]. FGSM [16], PGD [30], BIM [24], RGA attack proposed for SNN [6], AutoAttack [9] and common gaussian noise are used. The perturbations $\epsilon = 4/255$ (Table 3) and $\epsilon = 8/255$ (Table 4), step $k = 7$, and step size $\alpha = 0.01$. A fixed bandwidth of $\sigma = 0.4$ and 256 uniform bins within the range $[0, 255]$ are used in KDE-based MI calculation for all datasets. We set $\lambda = 0.05$ in Eq. (17), the sensitivity to $\lambda$ and more experimental settings can be found in Appendix.




(a) MI based on **temporal characteristics** of normal **direct training methods**

(b) MI based on **temporal characteristics** of **adversarial training methods**

(c) MI based on **spatial features** of normal **direct training methods**

(d) MI based on **spatial features** of **adversarial training methods**

Figure 3. Using MI based on the temporal characteristics (a)(b) or spatial features (c)(d) of SNN to characterize the robustness. Symbol shape denotes the network architecture (upper triangle: VGG11, lower triangle: AlexNet, circle: WideResNet16), color denotes the training method (red: STBP, green: TET, blue: RAT), and padding denotes the dataset (solid: CIFAR10, hollow: CIFAR100). Trends circled in yellow ovals represent the higher the mutual information, the lower the robustness when attacked by PGD [30], and vice versa, circled in gray ovals.

## 5.2. Necessity of Temporal Characteristics in SNNs for Robustness

**Using mutual information based on temporal characteristics to characterize the inherent robustness.** As shown in Fig. 3 (a), we find that, when SNNs under different direct training strategies (i.e., STBP [42] with red color, and TET [1] with green color), the higher the mutual information of the temporal characteristics, the worse the robustness of model when attacked by PGD (trends are circled in yellow ovals), no matter on CIFAR10 and CIFAR100 datasets with VGG11, AlexNet, and WideResNet16 network architectures. Furthermore, the same trends can also be found under adversarial training strategy as shown in Fig. 3 (b). These phenomenons show that mutual information of temporal characteristics of SNN can be used to characterize the robustness of model, and SNN with high mutual information between the input and the latent temporal characteristics is more vulnerable to adversarial attack. Then, to further evaluate the necessity of temporal characteristics for the robustness of SNN, we show the relationship between the mutual information based on the spatial feature of SNN and the robustness in Fig. 3 (c) and (d). The spatial feature is produced by the rate output of SNN, and the mutual information is calculated by MINE [4] proposed for ANN, which approximates the mutual information based on the batch samples. It can be seen from the Fig. 3 (c) and (d) that for different datasets and different networks, the relationship that the lower the mutual information, the higher the robustness, is not necessarily true (trends are circled in gray ovals). These phenomenons show that the mutual information obtained based on SNN spatial feature cannot be used to characterize the robustness of the model, which illustrates the necessity of the temporal characteristics of SNN to its robustness.

**Comparison with spatial feature-based MINE in ANNs.** From Table 1 we can observe, based on temporal characteristics of SNN, our TMI regularizer can bring more robustness improvements to SNN than the MINE regularizer [4] based on spatial feature in ANN. For example, on CIFAR-100, when attacked by FGSM, our TMI regularizer can bring ˜1.69% robustness improvement while MINE is ˜0.43%. This proves the necessity of using temporal characteristics to improve the robustness of the model.

**Temporal characteristics validation by time averaged spiking activity.** We use time averaged spiking activity (TASA) denoted in [23] to examine the spiking activity difference of SNN on original images and adversarial images. Specifically, the spiking activity (SA) of a layer is denoted as the ratio of the number of spikes produced over all the neurons accumulated across all time units $T$ of a layer to the total number of neurons present in that layer. And the TASA of a layer is denoted as the layer's SA divided by $T$. We calculate the TASA difference between the original image and the adversarial image of every layer on AlexNet with (STBP-TMI) or without (STBP) TMI regularizer. As recorded in Table 2, we find that without adding TMI regularizer (STBP), the TASA difference of AlexNet increases layer by layer, indicating that the perturbation information is more pronounced in the higher dimension than in the low dimension as illustrated in [38]. And as shown in the "STBP-TMI", we can observe that after reducing the mutual information of the temporal characteristics between the first layer and the last spike neuron layer (*i.e.*, the fifth layer of AlexNet), the TASA difference of the latent layer is maintained at a stable level (about 0.1240), and the TASA difference of the last layer is significantly reduced, which indicates the model learns more similar temporal features between original images and adversarial images after training with the TMI regularizer, enhancing inherent robustness to adversarial attack.

## 5.3. Overall Performance for Various Attack Types

We demonstrate effectiveness of our method across multiple datasets and training strategies under white-box, black-box adversarial attacks, and Gaussian noise. Results on CIFAR-100 are reported in Table 3, while results on CIFAR-10, DVS-CIFAR10, and Tiny-ImageNet are provided in the Appendix. We can observe, compared with the original

Table 1. The robustness improvement ($\Delta$) comparison between temporal characteristics-based TMI regularizer with others (*i.e.*, static data-based MINE proposed in ANNs and temporal characteristics-based AIMIE in thalamocortical network). The networks of SNN and ANN are VGGSNN and its ANN variant, with ReLU activation replacing LIF neurons.

| Datasets | Networks | Methods | $\Delta$ Natural | $\Delta$ FGSM | $\Delta$ PGD |
|---|---|---|---|---|---|
| CIFAR-10 | ANN | CLS-MINE | **0.03** | -0.25 | 0.07 |
| | SNN | CLS-AIMIE | -0.12 | 0.16 | 0.01 |
| | | CLS-MI (Ours) | 0.58 | **0.70** | **0.74** |
| CIFAR-100 | ANN | CLS-MINE | **0.08** | 0.43 | -0.02 |
| | SNN | CLS-AIMIE | -0.03 | -0.97 | -0.58 |
| | | CLS-MI (Ours) | 0.01 | **1.69** | **0.40** |

Table 2. The *time averaged spike activity* (TASA) difference on the original and adversarial images of every spike neuron layer on AlexNet with (*i.e.*, STBP-TMI) or without (*i.e.*, STBP) TMI regularizer. The dataset is CIFAR10. The attack is PGD.

| Spike neuron layer | STBP | STBP-TMI |
|---|---|---|
| layer 1 | 0.1152 | 0.1251 |
| layer 2 | 0.1299 | 0.1236 |
| layer 3 | 0.1265 | 0.1244 |
| layer 4 | 0.1342 | 0.1230 |
| layer 5 | 0.1661 | **0.0802** |

method, our TMI regularizer can effectively improve the robustness of the model. Specifically, in Table 3, on VGGSNN, compared with STBP, the STBP-TMI have improved the inherent robustness of model up to 2% and 1%, when attacked by FGSM and PGD, respectively. Furthermore, when compared with STBP-H and STBP-AIMIE, STBP-TMI still performs better, it verifies the effectiveness of our mutual information calculation method.

### 5.4. Comparision with SOTA robust SNNs methods.

We compare our method with SOTA robust SNN methods in Table 4, (*i.e.*, StoG [15], DLIF [14], SNN-RAT [12], and FEEL-SNN [44]. We apply our TMI regularizer to SNN-RAT [12] under the same settings. From Table 4 we can observe, our TMI regularizer can be used to further improve the robustness of these robust SNN methods. Specifically, on CIFAR-100 with VGG11, when attacked by FGSM, SNN-RAT improves the robustness of the original model (i.e., vanilla) from 4.30% to 25.86%, our TMI regularizer can further help improve the robustness to 27.32%.

### 5.5. Effectiveness of Proposed TMI regularizer

In this part, we show that our TMI regularizer does indeed enhance robustness by reducing the mutual information on the temporal characteristics. Fig. 4 shows the changing of the MI and robustness of SNNs before and after our TMI regularizer training. As shown in Fig. 4, after embedding the TMI regularizer to VGGSNN (*e.g.*, STBP-MI), the mutual information of the model is lower than that without TMI regularizer (*e.g.*, STBP). Thence, the robustness of STBP-

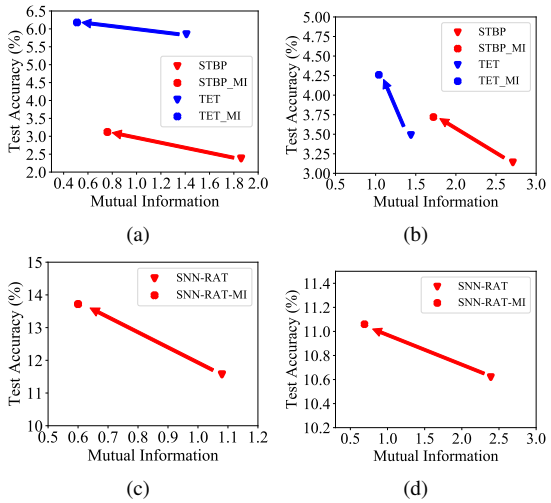



Figure 4. The changing of the MI and robustness of SNNs after TMI regularizer.

TMI has improved compared to STBP. To further validate the effectiveness of our mutual information calculation and implementation methods, we compare our approach with AIMIE [17], which employs a histogram partitioning technique to compute mutual information. Additionally, we assess whether the robustness improvement introduced by the TMI regularizer is due to genuine robustness rather than the phenomenon of "obfuscated gradients" [3]. Detailed analyses of both aspects are provided in the Appendix.

### 5.6. Ablation Study

To better understand TMI regularizer, we compare the performance with two different alternatives in our TMI regularizer implementation: (1) using the max-pooling strategy to replace the averaging strategy, (2) TMI by shallow features.

**Max-pooling strategy vs. average strategy.** As shown in Table 5, the robustness of model is reduced when taking the max-pooling strategy in the channel dimension. *e.g.*, when attacked by FGSM, the test accuracy of the "max-pooling" is 12.23% while our TMI regularizer is 25.80%. This phenomenon illustrates the effectiveness of averaging over channels, and shows that converting discrete spikes to continuous signals lead to less information loss.

**Shallow layers vs. deep layers.** Recalling $I(X;Z)$ denoted in Eq. (17), the latent representation $Z$ is obtained on the last spike neuron layer. We also conduct experiments on the temporal feature $Z'$ obtained on the latent spike neuron, *i.e.*, the fourth spike neuron layer in VGGSNN. The experimental results are recorded in Table 5. As can be observed, TMI regularizer with $I(X;Z)$ achieves the higher robustness of 4.36% on PGD and 25.80% on FGSM, which improves $\sim$ 3% over adopting $I(X;Z')$ in shallow layer. This illustrates that smaller mutual information can more accurately represent the extraction of input information.

Table 3. Test accuracy comparison of SNN models with/without TMI regularizer on **CIFAR-100 dataset**. "-**TMI**" represents embedding our proposed TMI regularizer into training, "-**H**" represents using output entropy [49], "-**AIMIE**" represents calculating mutual information by histogram method [17].

| Networks | Methods | Natural | white-box | | | | black-box | | | | Gaussian-noise |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | FGSM | FGSM-RGA | PGD | PGD-RGA | FGSM | FGSM-RGA | PGD | PGD-RGA | |
| VGGSNN | STBP | 68.11 | 13.51 | 13.78 | 2.05 | 2.72 | 31.70 | 31.34 | 24.60 | 24.61 | 18.98 |
| | STBP-H | **68.62** | 12.14 | 12.08 | 1.69 | 1.74 | 30.89 | 30.57 | 23.47 | 23.39 | 17.42 |
| | STBP-AIMIE | 67.98 | 12.54 | 12.28 | 1.47 | 1.34 | 31.67 | 31.59 | 25.04 | 25.01 | 17.67 |
| | **STBP-TMI(Ours)** | 68.12 | **15.20** | **15.21** | **2.45** | **3.17** | **32.33** | **32.20** | **27.83** | **26.38** | **19.25** |
| | TET | **72.66** | 15.19 | 15.78 | 2.25 | 3.05 | 35.78 | 35.63 | 28.04 | 28.13 | 19.41 |
| | TET-H | 71.80 | 14.67 | 14.55 | 2.08 | 2.10 | 34.98 | 34.87 | 28.13 | 28.05 | 17.28 |
| | TET-AIMIE | 71.89 | 14.93 | 15.52 | 2.11 | 2.46 | 35.85 | 35.25 | 28.10 | 28.44 | 16.13 |
| | **TET-TMI(Ours)** | 72.32 | **17.29** | **17.10** | **4.63** | **3.57** | **37.89** | **37.17** | **29.45** | **29.04** | **19.51** |
| AlexNet | STBP | 66.33 | 13.11 | 13.09 | 2.40 | 3.42 | 33.91 | 33.77 | 24.22 | 24.38 | 14.80 |
| | STBP-H | 65.78 | 8.76 | 8.55 | 2.18 | 2.23 | 32.90 | 31.78 | 23.04 | 23.12 | 13.90 |
| | STBP-AIMIE | 65.97 | 8.86 | 9.52 | 2.38 | 3.15 | 33.34 | 32.07 | 23.68 | 23.95 | 13.34 |
| | **STBP-TMI(Ours)** | **66.44** | **15.20** | **14.17** | **5.02** | **4.67** | **36.38** | **35.67** | **27.69** | **26.56** | **15.85** |
| | TET | 70.62 | 13.30 | 13.67 | 2.18 | 3.06 | 33.92 | 33.68 | 24.44 | 24.45 | 13.53 |
| | TET-H | 69.29 | 13.11 | 13.10 | 2.09 | 2.07 | 32.47 | 31.46 | 24.10 | 24.13 | 13.57 |
| | TET-AIMIE | 69.68 | 13.54 | 13.95 | 2.45 | 3.02 | 33.91 | 32.94 | 24.45 | 24.38 | 13.50 |
| | **TET-TMI(Ours)** | **71.02** | **16.20** | **15.03** | **4.67** | **4.56** | **35.51** | **34.23** | **26.27** | **25.89** | **16.89** |

Table 4. Comparison with **SOTA robust SNN methods**.

| Settings | Methods | Natural | FGSM | PGD | BIM | AA |
|---|---|---|---|---|---|---|
| CIFAR-10 VGG11, $T=4$ | vanilla | **92.42** | 12.07 | 0.35 | 0.23 | 0.10 |
| | **SNN-TMI** | 91.63 | **27.35** | **26.87** | **20.45** | **8.96** |
| CIFAR-10 VGG11 $T=8$ | vanilla | 91.81 | 11.23 | 0.32 | 0.20 | 0.06 |
| | StoG [15] | 91.64 | 16.22 | 0.28 | - | - |
| | DLIF [14] | 92.22 | 13.24 | 0.09 | - | - |
| | SNN-RAT [12] | 90.74 | 45.23 | 21.16 | 18.64 | 8.64 |
| | **SNN-TMI** | **92.58** | **49.53** | **25.86** | **22.14** | **10.45** |
| DVSCIFAR10 VGG11, $T=10$ | vanilla | **72.34** | 17.23 | 11.20 | 10.89 | 0.45 |
| | **SNN-TMI** | 70.23 | **30.12** | **25.12** | **30.62** | **7.45** |
| CIFAR-100 WR16, $T=8$ | vanilla | 72.51 | 5.32 | 0.02 | 0.03 | 0.01 |
| | **SNN-TMI** | **72.84** | **26.13** | **10.45** | **11.12** | **7.34** |
| CIFAR-100 VGG11 $T=8$ | vanilla | 72.76 | 4.30 | 0.09 | 0.08 | 0.07 |
| | StoG [15] | 70.44 | 8.27 | 0.49 | - | - |
| | DLIF [14] | 70.79 | 6.95 | 0.08 | - | - |
| | FEEL-SNN [44] | **73.79** | 9.60 | 2.04 | 1.81 | - |
| | SNN-RAT [12] | 70.89 | 25.86 | 10.38 | 9.50 | 5.73 |
| | FEEL-AT [44] | 69.79 | 18.67 | 11.07 | 10.56 | 5.89 |
| | **SNN-TMI** | 72.83 | **27.32** | **12.46** | **11.83** | **8.24** |
| ImageNet RN19, $T=4$ | vanilla | **42.89** | 6.77 | 2.78 | 2.54 | 1.03 |
| | **SNN-TMI** | 41.23 | **9.12** | **8.56** | **5.23** | **2.32** |

Table 5. Comparison between TMI regularizer with two different alternatives: (1) using max-pooling strategy, (2) MI by shallow features. The original accuracy (%) is described in the column "Natural". The dataset is CIFAR-10 with VGGSNN.

| | Natural | FGSM | PGD |
|---|---|---|---|
| Max-pooling strategy | 91.70 | 12.23 | 0.88 |
| MI by shallow feature | 92.00 | 22.60 | 2.45 |
| **Our TMI regularizer** | **92.39** | **25.80** | **4.36** |

## 5.7. Computational Cost

We have now reported the training cost in Table 6, where all results are averaged over 300 epochs on CIFAR100 with $T = 8$ and the VGG11 backbone. As shown, although our SNN-TMI introduces a mutual information regularizer, its training time is comparable to other regularization-based methods (e.g., DLIF and StoG) and is substantially lower than adversarial training approaches (AT and RAT). Specifically, SNN-TMI achieves an average training time of $160s$, which is only higher than Vanilla ($108s$), but significantly more efficient than AT ($300s$) and RAT ($294s$). This demonstrates that our KDE-based temporal MI estimation does not lead to prohibitive computational overhead. On the contrary, SNN-TMI achieves a favorable balance between efficiency and robustness.

Table 6. Training time of SNN-TMI. Results are averaged over 300 epochs on CIFAR100 with $T = 8$ and the VGG11 backbone. 's' represents the second.

| Method | Training time |
|---|---|
| Vanilla | $108s \pm 7s$ |
| FEEL-SNN [44] | $180s \pm 6s$ |
| DLIF [14] | $170s \pm 3s$ |
| StoG [15] | $150s \pm 3s$ |
| AT [38] | $300s \pm 6s$ |
| RAT [12] | $294s \pm 4s$ |
| **SNN-TMI (Ours)** | $160s \pm 6s$ |

## 6. Conclusion

In this paper, we revisit the robustness of SNNs from the perspective of temporal mutual information, highlighting the critical role of temporal characteristics in enhancing model robustness. Specifically, we first provide a theoretical analysis based on the information bottleneck principle, showing that the robustness error of a model is upper-bounded by the mutual information between the input features and the latent representations. We then demonstrate that the inherent temporal characteristics of SNNs naturally promote a more compact mutual information by cumulative firing mechanism and spike train dependencies analysis, thereby providing a tighter bound on the robustness error. Building upon this insight, we propose a novel temporal mutual information regularizer to strengthen the model's robustness. Extensive experiments on CIFAR-10, CIFAR-100, DVS-CIFAR10, Tiny-ImageNet, and ImageNet confirm that temporal characteristics are crucial for improving SNN robustness. Moreover, our proposed method consistently enhances the inherent robustness of various SNN architectures across different types of adversarial attacks and noise conditions.

## 7. Acknowledgements

This work was supported in part by the STI 2030 Major Projects under Grant 2021ZD0200400, in part by the National Natural Science Foundation of China (62376247, 62436008, 62334014), in part by the grants from Key R&D Program of Zhejiang (2022C01048), in part by Fundamental Research Funds for the Central Universities (226-2025-00057), and in part by Chongqing Natural Science Foundation Innovation and Development Joint Fund (No. CSTB2025NSCQ-LZX0160).

## References

[1] Alessandro Achille and Stefano Soatto. Information dropout: Learning optimal representations through noisy computation. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2897–2905, 2018. 6

[2] Rana Ali Amjad and Bernhard C Geiger. Learning representations for neural network-based classification using the information bottleneck principle. *IEEE transactions on pattern analysis and machine intelligence*, 42(9):2225–2239, 2019. 2, 3, 4

[3] Anish Athalye, Nicholas Carlini, and David Wagner. Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In *Proc. of International Conference on Machine Learning*, pages 274–283. PMLR, 2018. 7

[4] Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeshwar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and Devon Hjelm. Mutual information neural estimation. In *Proc. of International Conference on Machine Learning*, pages 531–540. PMLR, 2018. 3, 6

[5] Abhiroop Bhattacharjee, Ruokai Yin, Abhishek Moitra, and Priyadarshini Panda. Are SNNs truly energy-efficient?—a hardware perspective. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 13311–13315. IEEE, 2024. 1

[6] Tong Bu, Jianhao Ding, Zecheng Hao, and Zhaofei Yu. Rate gradient approximation attack threats deep spiking neural networks. In *Proc. of Computer Vision and Pattern Recognition*, pages 7896–7906. IEEE, 2023. 5

[7] Hojae Choi, Jaewook Kim, Jongkil Park, Seongsik Park, Hyun Jae Jang, Seung Hwan Lee, Byeong-Kwon Ju, and YeonJoo Jeong. STAR-SNN: A spatio-temporal adaptive recurrent spiking neural network with separated propagation surrogate gradient for hardware efficient real-time learning. *Neurocomputing*, page 132968, 2026. 1

[8] Thomas M Cover. *Elements of information theory*. John Wiley & Sons, 1999. 3

[9] Francesco Croce and Matthias Hein. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In *Proc. of International Conference on Machine Learning*, pages 2206–2216. PMLR, 2020. 5

[10] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *Proc. of Computer Vision and Pattern Recognition*, pages 248–255. IEEE, 2009. 5

[11] Shikuang Deng, Yuhang Li, Shanghang Zhang, and Shi Gu. Temporal efficient training of spiking neural network via gradient re-weighting. In *International Conference on Learning Representations*, 2022. 5

[12] Jianhao Ding, Tong Bu, Zhaofei Yu, Tiejun Huang, and Jian Liu. SNN-RAT: Robustness-enhanced spiking neural network through regularized adversarial training. *Advances in Neural Information Processing Systems*, 35:24780–24793, 2022. 1, 2, 7, 8

[13] Jianhao Ding, Zhaofei Yu, Tiejun Huang, and Jian K Liu. Spike timing reshapes robustness against attacks in spiking neural networks. *arXiv preprint arXiv:2306.05654*, 2023. 2

[14] Jianhao Ding, Zhiyu Pan, Yujia Liu, Zhaofei Yu, and Tiejun Huang. Robust stable spiking neural networks. In *Proc. of International Conference on Machine Learning*, pages 11016–11029. PMLR, 2024. 1, 2, 7, 8

[15] Jianhao Ding, Zhaofei Yu, Tiejun Huang, and Jian K Liu. Enhancing the robustness of spiking neural networks with stochastic gating mechanisms. In *AAAI Conference on Artificial Intelligence*, pages 492–502, 2024. 7, 8

[16] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014. 5

[17] Ekaterina D Gribkova, Baher A Ibrahim, and Daniel A Llano. A novel mutual information estimator to measure spike train correlations in a model thalamocortical network. *Journal of neurophysiology*, 120(6):2730–2744, 2018. 4, 7, 8

[18] Yufei Guo, Yuanpei Chen, Liwen Zhang, Xiaode Liu, Yinglei Wang, Xuhui Huang, and Zhe Ma. Im-loss: information maximization loss for spiking neural networks. *Advances in Neural Information Processing Systems*, 35:156–166, 2022. 1

[19] Yufei Guo, Xiaode Liu, Yuanpei Chen, Weihang Peng, Yuhan Zhang, and Zhe Ma. Spiking transformer: Introducing accurate addition-only spiking self-attention for transformer. In *Proc. of Computer Vision and Pattern Recognition*, pages 24398–24408. IEEE, 2025.

[20] Yangfan Hu, Qian Zheng, Xudong Jiang, and Gang Pan. Fast-SNN: Fast spiking neural network by converting quantized ANN. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(12):14546–14562, 2023. 1

[21] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 5

[22] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25, 2012. 5

[23] Souvik Kundu, Massoud Pedram, and Peter A Beerel. Hire-snn: Harnessing the inherent robustness of energy-efficient deep spiking neural networks by training with crafted input noise. In *Proc. of International Conference on Computer Vision*, pages 5209–5218, 2021. 1, 2, 6

[24] Alexey Kurakin, Ian J Goodfellow, and Samy Bengio. Adversarial examples in the physical world. In *Artificial intelligence safety and security*, pages 99–112. Chapman and Hall/CRC, 2018. 5

[25] Hongmin Li, Hanchao Liu, Xiangyang Ji, Guoqi Li, and Luping Shi. Cifar10-DVS: an event-stream dataset for object classification. *Frontiers in neuroscience*, 11:244131, 2017. 5

[26] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information theory*, 37(1):145–151, 2002. 3

[27] Yujia Liu, Tong Bu, Jianhao Ding, Zecheng Hao, Tiejun Huang, and Zhaofei Yu. Enhancing adversarial robustness in snns with sparse gradients. In *Proc. of International Conference on Machine Learning*, pages 30738–30754. PMLR, 2024. 1

[28] De Ma, Xiaofei Jin, Shichun Sun, Yitao Li, Xundong Wu, Youneng Hu, Fangchao Yang, Huajin Tang, Xiaolei Zhu, Peng Lin, et al. Darwin3: a large-scale neuromorphic chip with a novel isa and on-chip learning. *National Science Review*, 11(5):nwae102, 2024. 1

[29] Jordi Madrenas, Bernardo Vallejo-Mancero, Josep Àngel Oltra-Oltra, Mireya Zapata, Jordi Cosp-Vilella, Robert Calatayud, Satoshi Moriya, and Shigeo Sato. Real-time adaptive physical sensor processing with snn hardware. In *International Conference on Artificial Neural Networks*, pages 423–434. Springer, 2023. 1

[30] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations*, 2018. 1, 5, 6

[31] Young-Il Moon, Balaji Rajagopalan, and Upmanu Lall. Estimation of mutual information using kernel density estimators. *Physical Review E*, 52(3):2318, 1995. 4, 5

[32] Ben Poole, Sherjil Ozair, Aaron Van Den Oord, Alex Alemi, and George Tucker. On variational bounds of mutual information. In *Proc. of International Conference on Machine Learning*, pages 5171–5180. PMLR, 2019. 3

[33] Ohad Shamir, Sivan Sabato, and Naftali Tishby. Learning and generalization with the information bottleneck. *Theoretical Computer Science*, 411(29-30):2696–2711, 2010. 3

[34] Saima Sharmin, Priyadarshini Panda, Syed Shakib Sarwar, Chankyu Lee, Wachirawit Ponghiran, and Kaushik Roy. A comprehensive analysis on adversarial robustness of spiking neural networks. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2019. 1

[35] Saima Sharmin, Nitin Rathi, Priyadarshini Panda, and Kaushik Roy. Inherent adversarial robustness of deep spiking neural networks: Effects of discrete input encoding and nonlinear activations. In *European Conference on Computer Vision*, pages 399–414. Springer, 2020. 1

[36] Ali Siddique, Mang I Vai, and Sio Hang Pun. A low cost neuromorphic learning engine based on a high performance supervised snn learning algorithm. *Scientific Reports*, 13(1): 6280, 2023. 1

[37] Ajay Subramanian, Elena Sizikova, Najib Majaj, and Denis Pelli. Spatial-frequency channels, shape bias, and adversarial robustness. *Advances in neural information processing systems*, 36:4137–4149, 2023. 2

[38] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks. In *International Conference on Learning Representations*, 2014. 2, 6, 8

[39] Naftali Tishby and Noga Zaslavsky. Deep learning and the information bottleneck principle. In *IEEE information theory workshop (itw)*, pages 1–5. IEEE, 2015. 1, 2

[40] Naftali Tishby, Fernando C Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000. 2

[41] Tim Van Erven and Peter Harremos. Rényi divergence and kullback-leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014. 2

[42] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in neuroscience*, 12:331, 2018. 6

[43] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, Yuan Xie, and Luping Shi. Direct training for spiking neural networks: Faster, larger, better. In *AAAI Conference on Artificial Intelligence*, pages 1311–1318, 2019. 1, 2

[44] Mengting Xu, De Ma, Huajin Tang, Qian Zheng, and Gang Pan. FEEL-SNN: Robust spiking neural networks with frequency encoding and evolutionary leak factor. *Advances in Neural Information Processing Systems*, 37:91930–91950, 2024. 1, 2, 7, 8

[45] Jiaqi Yan, Changping Wang, De Ma, Huajin Tang, Qian Zheng, and Gang Pan. Training high performance spiking neural network by temporal model calibration. In *Proc. of International Conference on Machine Learning*. PMLR, 2025. 1

[46] Jian Zhang, Yong Wang, Yimao Cai, Bo Bi, Qiliang Chen, and Yanlong Zhang. A low power spiking neural network accelerator on fpga for real-time edge computing. In *2025 Joint International Conference on Automation-Intelligence-Safety (ICAIS) & International Symposium on Autonomous Systems (ISAS)*, pages 1–6. IEEE, 2025. 1

[47] Xingyu Zhang, Yu Cao, Jian Huang, Jindong Liu, and Zhi-Qiang Zhang. A systematic review of spiking neural networks for human-robot interaction in rehabilitative wearable robotics. *IEEE Transactions on Cognitive and Developmental Systems*, 2025. 1

[48] Long Zhao, Ting Liu, Xi Peng, and Dimitris Metaxas. Maximum-entropy adversarial data augmentation for improved generalization and robustness. *Advances in Neural Information Processing Systems*, 33:14435–14447, 2020. 3, 4

[49] Dawei Zhou, Nannan Wang, Xinbo Gao, Bo Han, Xiaoyu Wang, Yibing Zhan, and Tongliang Liu. Improving adversarial robustness via mutual information estimation. In *Proc. of International Conference on Machine Learning*, pages 27338–27352. PMLR, 2022. 8